STUDENT PERFORMANCE DATA ANALYSIS USING EXCEL

APRIL 16, 2026

RAFIQUE AHMED MEMON

Contents

Title: Student Performance Data Analysis Using Excel 2

Introduction 2

Methodology 2

Data Analysis 2

Key Findings 3

Conclusion 3

References 3

Appendix 4

Title: Student Performance Data Analysis Using Excel

Introduction

This project focuses on analyzing student performance data using Microsoft Excel. The aim of the analysis is to identify patterns between study hours and exam scores, evaluate overall performance, and generate insights that can support academic improvement (Meyers, 2013).

Methodology

The dataset consists of 10 students, including variables such as Student ID, study hours, and exam scores. A new variable, "Performance," was created using the IF function to classify students as "High Performer" or "Average" based on their exam scores.

The analysis was conducted using Excel functions and tools, including:

- Descriptive statistics (mean, minimum, maximum)
- Conditional functions (IF, COUNTIF)
- Correlation analysis
- Data visualization (charts)
- Pivot tables for summarization

Data Analysis

Descriptive Statistics

The average score of students was calculated to understand overall performance, along with identifying the highest and lowest scores.

Performance Classification

Students scoring above 80 were classified as "High Performers," while others were labelled as "Average." This helped in segmenting the dataset for deeper analysis.

Correlation Analysis

A correlation analysis was performed between study hours and exam scores using the CORREL function. The results indicated a positive relationship, suggesting that increased study time is associated with better academic performance (Field, 2018).

Data Visualization

Various charts were created to present the findings effectively:

- Bar chart to compare student scores
- Pie chart to show performance distribution
- Scatter plot to analyses the relationship between study hours and scores

Pivot Table Analysis

Pivot tables were used to summarize the number of students in each performance category and calculate the average score for each group.

Key Findings

- Students who studied more hours generally achieved higher scores.
- A significant proportion of students were classified as high performers.
- There is a clear positive relationship between study hours and academic performance.

Conclusion

The analysis demonstrates that study habits have a strong impact on student performance. The use of Excel tools such as pivot tables, charts, and statistical functions enabled efficient data analysis and meaningful insights. This project highlights the importance of data-driven decision-making in educational and professional contexts (Elena Karamazova, 2017).

References

Elena Karamazova, T. J. Z. Z. T., 2017. Analysing and Comparing the Final Grade in Mathematics by Linear Regression Using Excel and SPSS. *International Journal of Mathematics Trends and Technology (IJMTT),* Volume 52, no. 5, pp. 334-344.

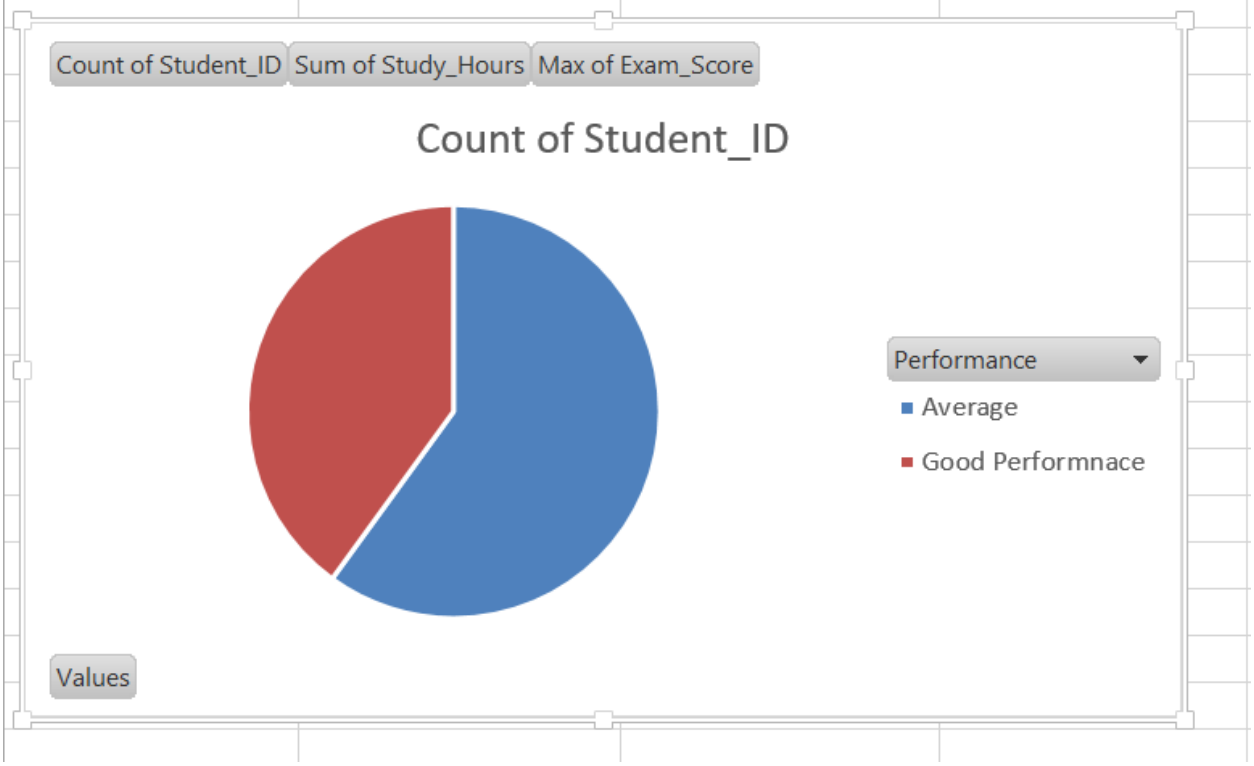
Field, A., 2018. *Discovering Staticstics using IBM SPSS.* 5th ed. London: Sage.

Meyers, L. G. G. a. G. A., 2013. Performing data analysis using IBM SPSS.

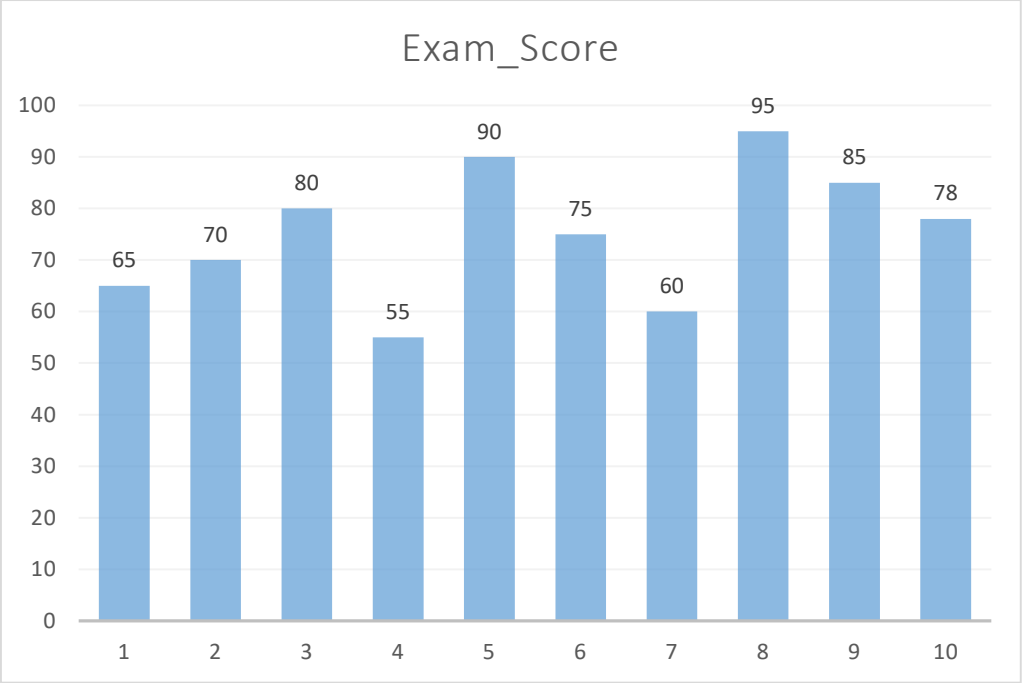
Appendix

Pivot table and Pie chart

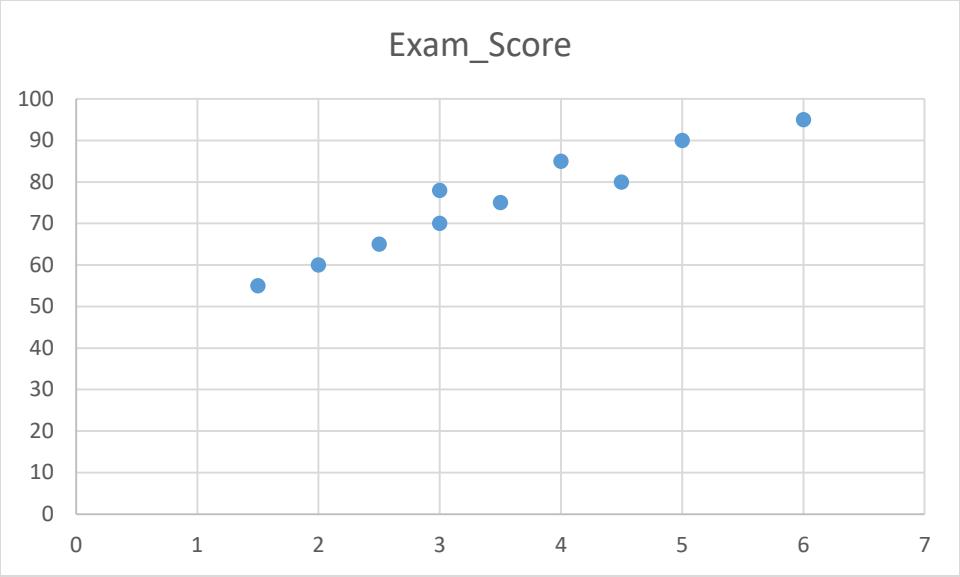
Row Labels	Count of Student_ID	Sum of Study_Hours	Max of Exam_Score
Average	6	15.5	78
Good Performnace	4	19.5	95
Grand Total	**10**	**35**	**95**



Bar Graph



Scatter plot


Exam_Score
100
90
80
70
60
50
40
30
20
10
0
0
1
2
3
4
5
6
7